

Assignment 1: Classical Machine Learning PART II – Report

Methods:

1. Gradient Boosting (with and without PCA) (**Friedman, 2001**)
2. KNN (with and without PCA) (**Cover and Hart, 1967**)
3. Random Forest (with and without PCA) (**Breiman, 2001**)
4. SVM (with and without PCA) (**Cortes and Vapnik, 1995**)

Method	Pipeline	Hyperparameters	Cross-validation
Gradient Boosting (with & without PCA)	NA	n_estimators [100, 200] learning_rate [0.05, 0.1] max_depth [2, 3]	cv = 3
K-Nearest Neighbors	StandardScaler → KNN	n_neighbors [3, 5, 7, 9, 11, 13, 15] weights [uniform, distance] metric [euclidean, manhattan]	cv = 5
PCA+K-Nearest Neighbors	StandardScaler → KNN	n_components: [5, 10, 15, 20, 25] n_neighbors [3, 5, 7, 9, 11, 13, 15] weights [uniform, distance] metric [euclidean, manhattan]	cv = 5
Random Forest (with & without PCA)	NA	n_estimators [100, 200] max_depth [None, 20] max_features [sqrt, log2] min_samples_split [2, 5]	cv = 5
Support Vector Machine (with & without PCA)	StandardScaler → SVC (probability=True)	C [0.1, 1, 10] kernel [rbf, linear] gamma [scale, auto]	cv = 5

Table 1: Hyperparameter settings for classification models

Results:

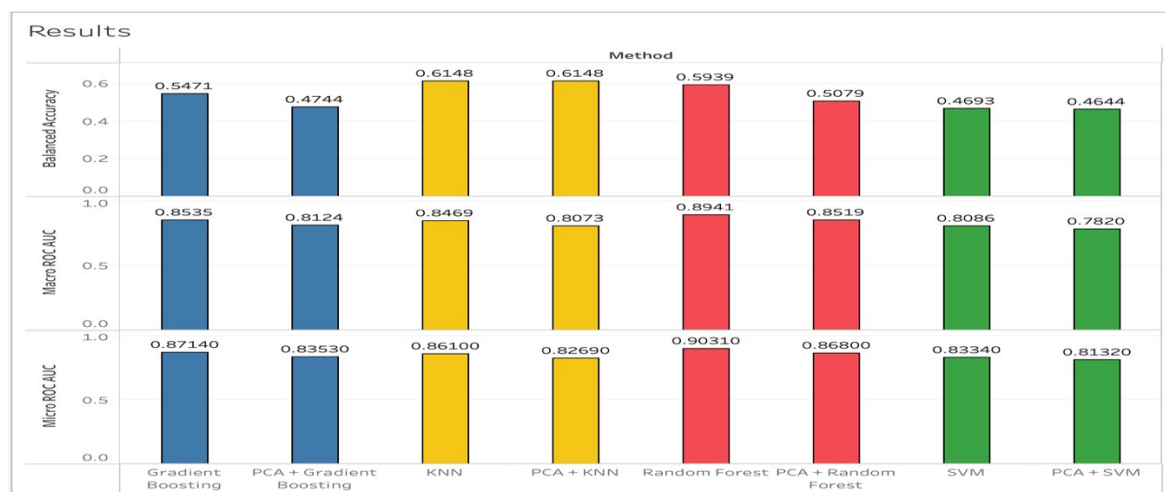


Figure 1. Model performance comparison across evaluation metrics

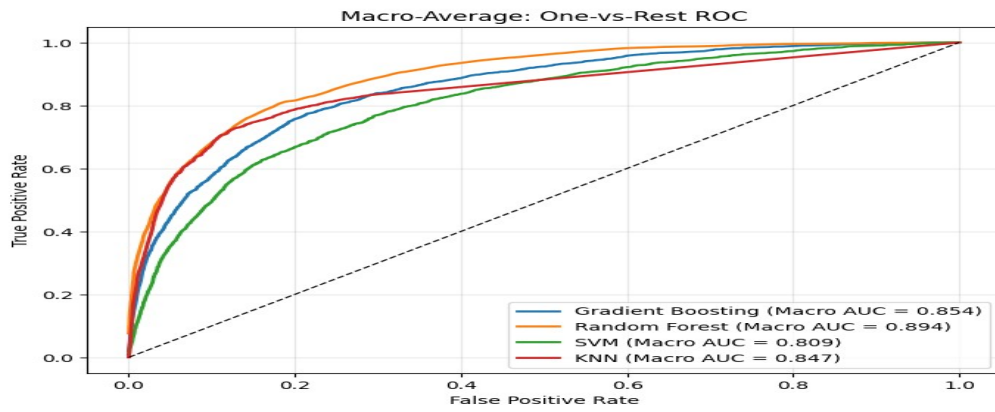


Figure 2. Macro-average One-vs-Rest ROC curves for baseline classification models

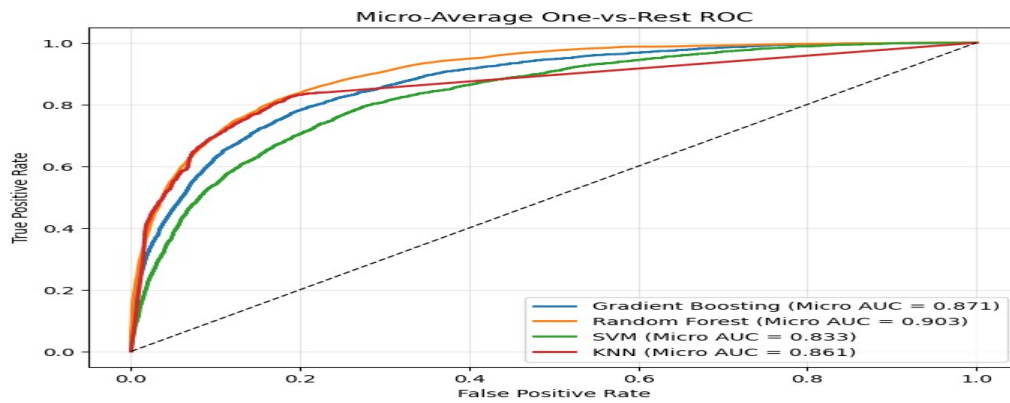


Figure 3. Micro-average One-vs-Rest ROC curves for baseline classification models

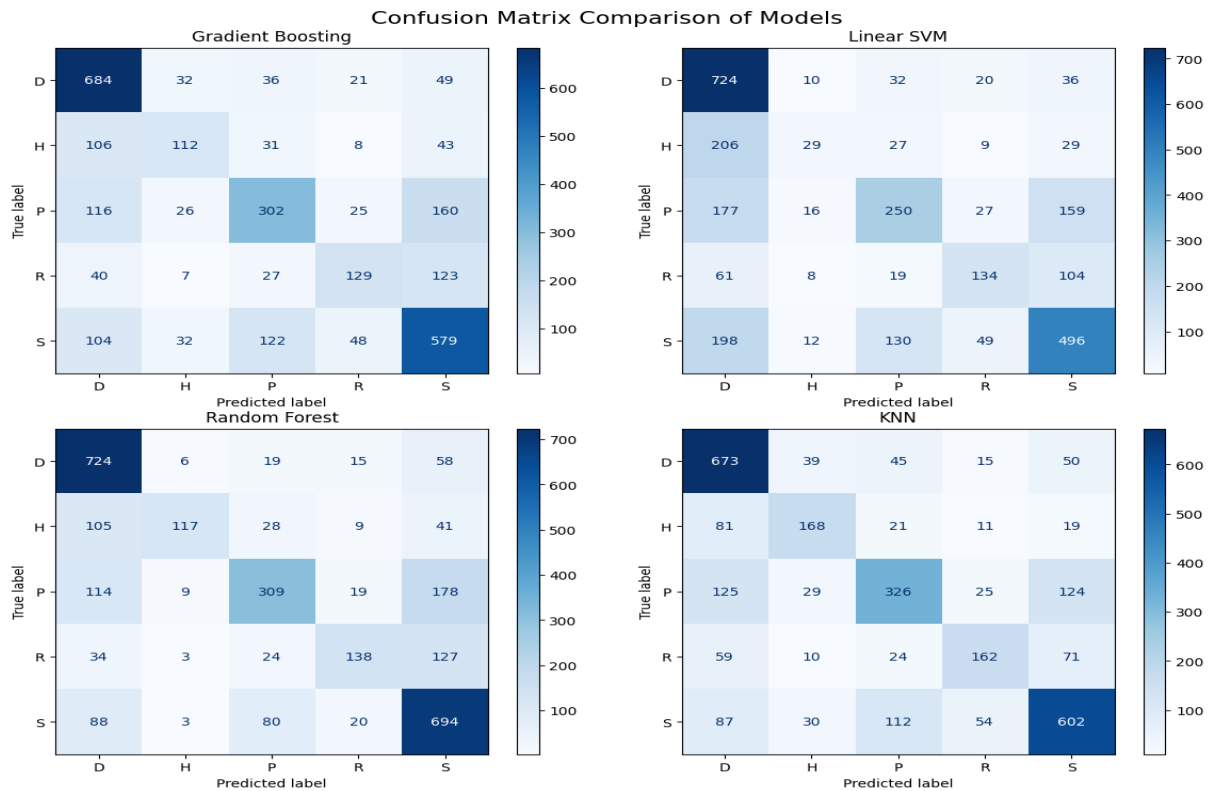


Figure 4. Confusion matrix comparison for baseline classification models

Conclusions:

1. Dimensionality reduction by PCA does not improve model performance on this data, as indicated by Figure 1. In all classifiers, the PCA-based variants generally have lower or equivalent balanced accuracy and lower ROC AUC than their respective baselines. This implies that the transformation is losing information. For instance, Random Forest decreases from 0.5939 to 0.5079 in balanced accuracy, and from 0.8941 to 0.8519 in macro ROC AUC after applying PCA. Likewise, the performance of Gradient Boosting also decreases from 0.5471 to 0.4744 in balanced accuracy, and SVM decreases from 0.4693 to 0.4644.
2. Based on Figures 2 and 3, Random Forest has the best performance as its ROC curves are closest to the top-left corner in both macro and micro-average figures. This aligns with its highest macro-ROC AUC and micro-ROC AUC values of 0.894 and 0.903, as shown in the figures.
3. KNN demonstrates improved performance on minority classes, as seen in Figure 4. It has a higher number of correctly classified samples for class H, which has 168 samples, and class R, which has 162 samples, in comparison to the number of correctly classified samples by the Random Forest and Gradient Boosting algorithms, which are 117 for class H and 138 for class R, and 112 for class H and 129 for class R, respectively.

*** Although Logistic Regression was included in the original model, it was removed and replaced by KNN in the final model. Any mention of Logistic Regression was not intended.

*** In the PCA + KNN grid search, the distance metric was mistakenly written as “Manhattan” instead of “manhattan”, which caused some parameter combinations to fail during cross-validation.

*** One minor implementation detail in the evaluation of PCA + KNN was identified (`classification_report(y_test, y_pred_knn)` instead of `classification_report(y_test, y_pred_pca_knn)`). The corrected results indicate that PCA + KNN has a lower performance compared to KNN, in line with the overall conclusion that dimensionality reduction does not improve performance.

*** In the submitted implementation, the balanced accuracy for PCA + KNN was initially computed using baseline KNN predictions due to an oversight (`balanced_accuracy_score(y_test, y_pred_knn)` instead of `balanced_accuracy_score(y_test, y_pred_pca_knn)`). The corrected results show a lower balanced accuracy for PCA + KNN.

References:

1. **Friedman, J.H. (2001).** *Greedy function approximation: A gradient boosting machine*. Annals of Statistics, 29(5), pp.1189–1232. Available at: <https://projecteuclid.org/journals/annals-of-statistics/volume-29/issue-5/Greedy-function-approximation-A-gradient-boosting-machine/10.1214/aos/1013203451.full>
2. **Cover, T. and Hart, P. (1967).** *Nearest neighbor pattern classification*. IEEE Transactions on Information Theory, 13(1), pp.21–27. Available at: <https://ieeexplore.ieee.org/document/1053964>
3. **Breiman, L. (2001).** *Random forests*. Machine Learning, 45(1), pp.5–32. Available at: <https://link.springer.com/article/10.1023/A:1010933404324>
4. **Cortes, C. and Vapnik, V. (1995).** *Support-vector networks*. Machine Learning, 20(3), pp.273–297. Available at: <https://link.springer.com/article/10.1007/BF00994018>

This submission is my own independent work and all sources used in the submission are referenced. Generative AI was USED for this submission, as outlined by the relevant references. For each reference, I provide a full record of prompts given to the generative AI tool, and responses received below. I have also included references that acknowledge the usage of AI in the reference list of this coursework, based on the guidance provided at: AI and Assessment: Digital Learning & Tools

Prompt 1:

suggest appropriate title for this confusion matrix

→

Here are **appropriate academic titles** for your confusion matrix figure

☐ **Best title (recommended)**

Figure X. Confusion matrices for baseline classification models

☐ **Slightly more descriptive (better for marks)**

Figure X. Comparison of confusion matrices across baseline classification models

☐ **More detailed version**

Figure X. Confusion matrices illustrating classification performance of Gradient Boosting, SVM, Random Forest, and KNN

☐ **My recommendation for YOU**

Use this one (best balance of clarity + professionalism)

Figure X. Comparison of confusion matrices across baseline classification models

□ **How to refer in text**

“As shown in Figure X, KNN and Random Forest exhibit stronger diagonal dominance compared to other models.”

If you want, I can:

Write a **high-scoring explanation paragraph for this figure**

Or align it perfectly with your **Results + Conclusion sections**

Prompt 2:

Provide an explanation of why KNN may perform better on minority classes.

→

Here is a **clear, academic explanation** you can use in your report

k-Nearest Neighbours (KNN) may perform better on minority classes because it makes predictions based on local neighbourhoods rather than global decision boundaries.

Unlike models such as Random Forest or SVM, which are influenced by overall data distribution, KNN considers the closest data points when assigning a class label. This allows it to better capture small or less frequent class patterns. As a result, minority class instances are more likely to be correctly classified if they form local clusters in the feature space.

However, this advantage depends on the quality of the feature representation and the choice of distance metric.

□ **Shorter version (if needed)**

KNN performs better on minority classes because it relies on local neighbourhood information rather than global patterns, allowing it to better capture smaller class clusters.

If you want, I can:

Link this directly with your **confusion matrix results (for higher marks)**

Or refine it to match your **report tone perfectly**